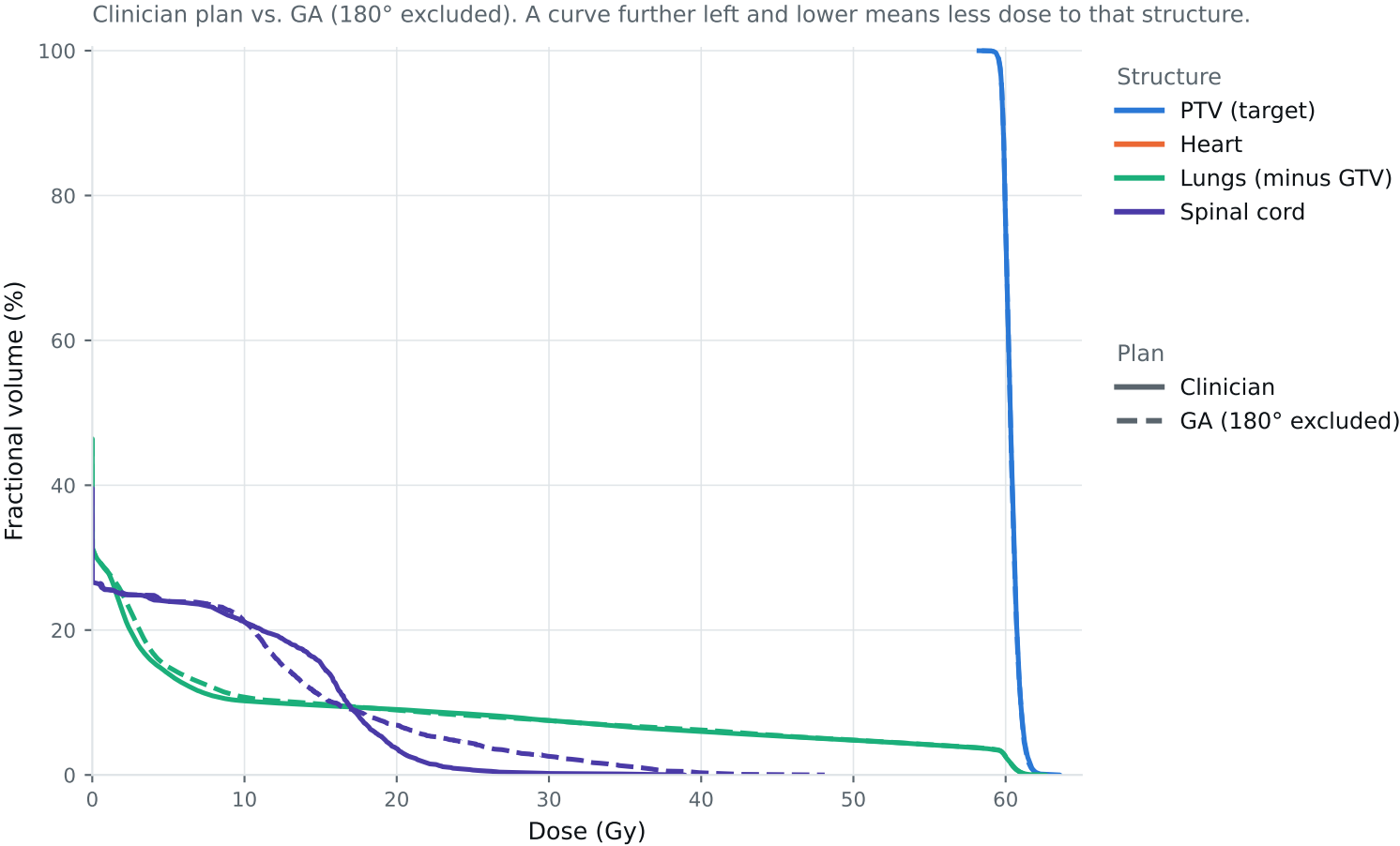


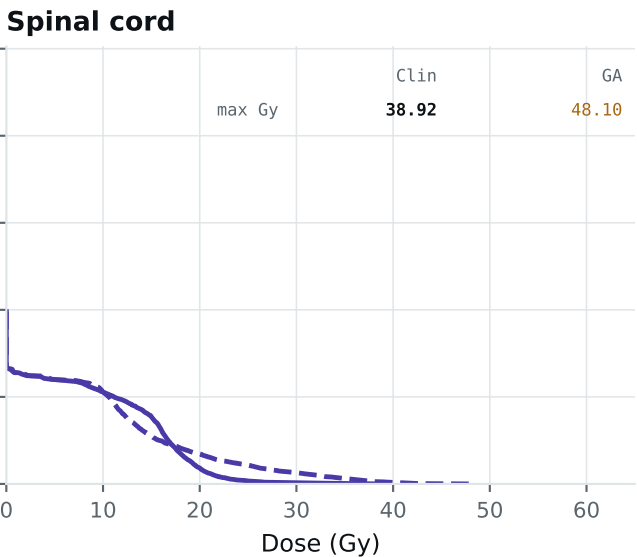
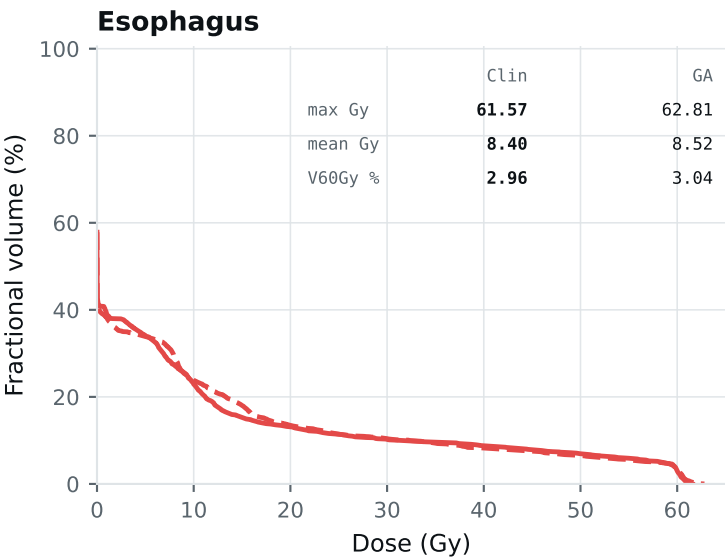
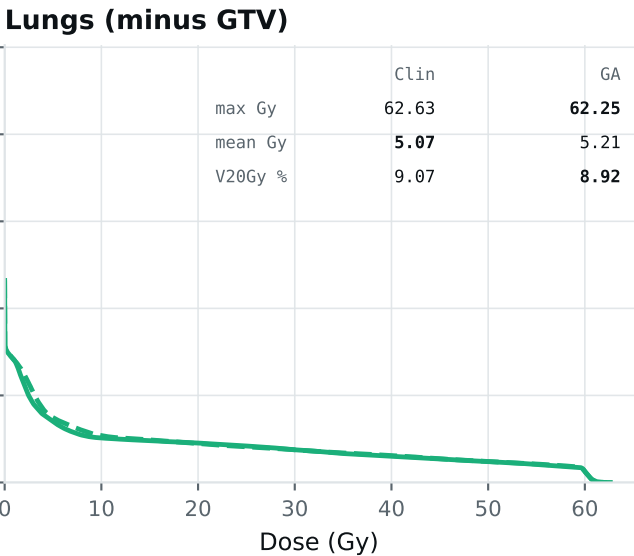
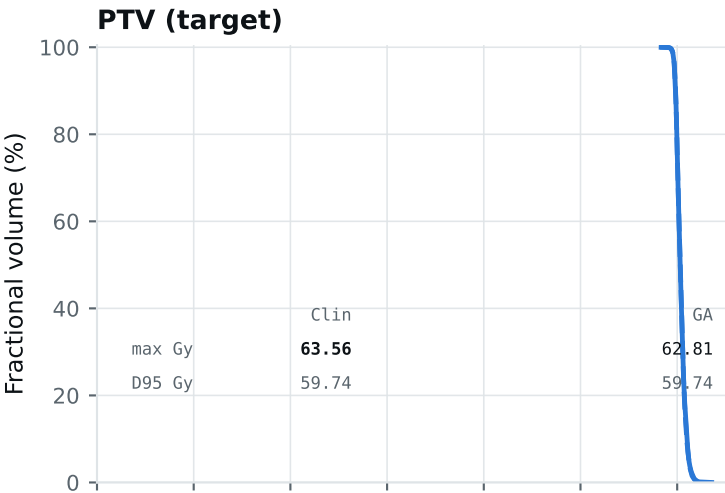
# Dose-volume histogram — Lung\_Patient\_7



Curves further left are better for organs; the PTV curve should stay high, then drop sharply. Target curves should overlap; organ curves show the tradeoffs. Esophagus is on page 2.

# DVH by structure — Lung\_Patient\_7

Same curves, one structure per panel, all plans, each annotated with that structure's protocol metrics.  
Bold = clinically preferable: organs favor lower dose; PTV favors the value closest to goal within the limit. Grey = all plans are within 0.01 of each other. Amber misses a goal,



Plan

- Clinician
- - GA (180° excluded)

# Clinical criteria — Lung\_Patient\_7

Organs: lower is better. PTV: closest to goal without exceeding the limit. Bold marks the best plan.

| Criterion                       | Limit     | Goal                                   | Clinician    | GA<br>180° excluded |    |
|---------------------------------|-----------|----------------------------------------|--------------|---------------------|----|
| GTV max                         | 69        | 66                                     | 62.26        | <b>61.93</b>        | Gy |
| PTV max                         | 69        | 66                                     | <b>63.56</b> | 62.81               | Gy |
| ESOPHAGUS max                   | 66        | —                                      | <b>61.57</b> | 62.81               | Gy |
| ESOPHAGUS mean                  | 34        | 21                                     | <b>8.40</b>  | 8.52                | Gy |
| ESOPHAGUS V60Gy                 | 17        | —                                      | <b>2.96</b>  | 3.04                | %  |
| HEART max                       | 66        | —                                      | 0.00         | 0.00                | Gy |
| HEART mean                      | 27        | 20                                     | 0.00         | 0.00                | Gy |
| HEART V30Gy                     | 50        | 48                                     | 0.00         | 0.00                | %  |
| LUNG_L max                      | 66        | —                                      | <b>14.84</b> | 21.61               | Gy |
| LUNG_R max                      | 66        | —                                      | 62.63        | <b>62.25</b>        | Gy |
| CORD max                        | 50        | 48                                     | <b>38.92</b> | 48.10               | Gy |
| SKIN max                        | 60        | —                                      | 45.00        | <b>44.43</b>        | Gy |
| LUNGS_NOT_GTV max               | 66        | —                                      | 62.63        | <b>62.25</b>        | Gy |
| LUNGS_NOT_GTV mean              | 21        | 20                                     | <b>5.07</b>  | 5.21                | Gy |
| LUNGS_NOT_GTV V20Gy             | 37        | —                                      | 9.07         | <b>8.92</b>         | %  |
| PTV D95 (target coverage)       |           |                                        | 59.74        | 59.74               |    |
| Objective value (search signal) |           |                                        | 66.92        | 67.81               |    |
| Solve time                      |           |                                        | 20 s         | 20 s                |    |
| Beam angles (gantry°)           | Clinician | 0°, 185°, 210°, 240°, 270°, 300°, 330° |              |                     |    |
|                                 | GA        | 15°, 45°, 165°, 185°, 255°, 285°, 315° |              |                     |    |

Grey rows: all plans agree within 0.01 — heart, lung and LUNGS\_NOT\_GTV max sit at 66 Gy because those structures abut the target. Amber misses the goal; red exceeds the limit.

Source: Lung\_Patient\_7\_clinical\_metrics\_full\_resolution.json, full resolution. The curves and table come from the same solve.